

Tropical cyclone - a giant heat engine

Y12 (2012) — English translation

This problem considers a simplified model of a tropical cyclone. In this model, the cyclone is considered as a giant heat engine operating according to the Carnot cycle.

Fig. 1. Diagram of air mass flows

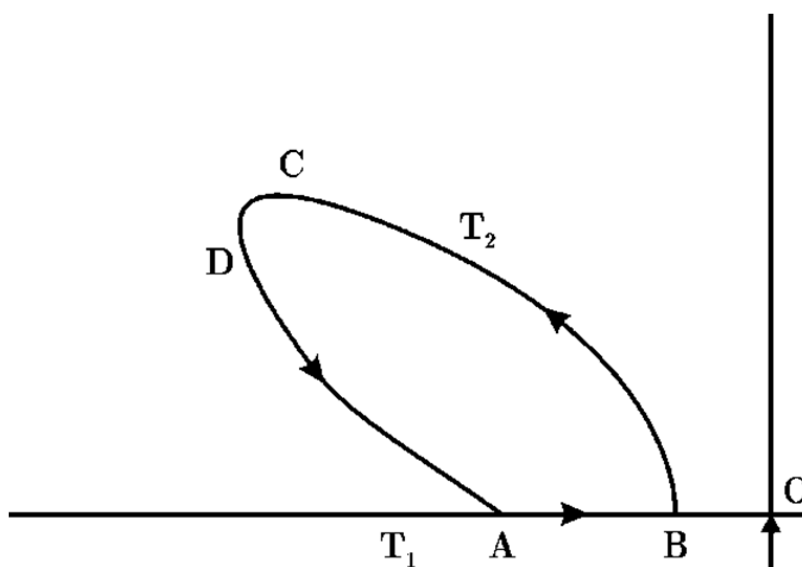


Figure 1: Figure 1.

Fig. 2. Carnot cycle

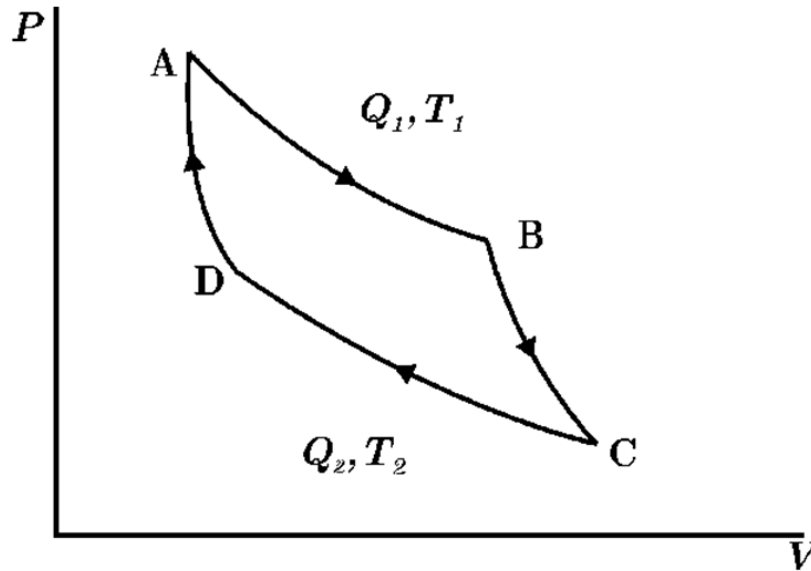


Figure 2: Figure 2.

A mass of dry air Δm flows along the ocean surface from a high pressure area (A) to a low pressure area (B) close to the center of the storm O (Fig. 1), capturing moisture (water vapor) along the way. Together with the vapors, heat Q_1 is taken from the ocean. Air masses saturated with water vapor rise into the troposphere (section BC). In the troposphere (section CD) they are discharged as a tropical shower, giving off energy Q_2 .

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|-----------|---|----------|
| A1 | Using the first law of thermodynamics, find an expression for Q_1 (see Fig. 2) in terms of L - the specific heat of evaporation, Δq - the moisture mass of the trapped air mass Δm on the way from A to B P_A and P_B - pressure in point A and point B respectively. Use the universal gas constant R , sea surface temperature T_1 , molar mass M of air. | 0 |
| A2 | Find the total work done in one Carnot cycle by a heat engine on a mass of air Δm . | 0 |
| A3 | Assuming that the work done by the heat engine goes entirely to increasing the kinetic energy of the air mass, obtain an expression for the speed v_B of the air in the central part of the cyclone (the "eye" of the cyclone). | 0 |

A4 Assuming $v_A = 0$, find the numerical value of v_B using the following data: **0**

- $T_1 = 273 \text{ K}$, $T_2 = 213 \text{ K}$;
- $R = 8.31 \text{ J}/(\text{mol} \cdot \text{K})$;
- relative air humidity at B $\varphi = 75\%$;
- at $T = T_1$ saturated vapor pressure $P_{sat} = 42 \text{ mbar}$;
- $P_A = 1000 \text{ mbar}$, $P_B = 950 \text{ mbar}$;
- average molar mass of air $M = 29 \cdot 10^{-3} \text{ kg} \cdot \text{mol}^{-1}$;
- molar mass of water $M_w = 18 \cdot 10^{-3} \text{ kg} \cdot \text{mol}^{-1}$;
- specific heat of evaporation of water $L = 2.26 \cdot 10^6 \text{ J} \cdot \text{kg}^{-1}$;
- density of dry air $\rho = 1.3 \text{ kg}/\text{m}^3$ ($1 \text{ bar} = 10^5 \text{ Pa}$).

A5 Assuming that the azimuthal velocity v and the radial distance r from the center of the cyclone are related by $v\sqrt{r} = \text{const}$, and that the velocity v_A at the outer edge of the cyclone is 10 m/s , calculate the effective radius of the cyclone (assume $r_B = 10 \text{ km}$). **0**

Source: <https://pho.rs/p/3971>